

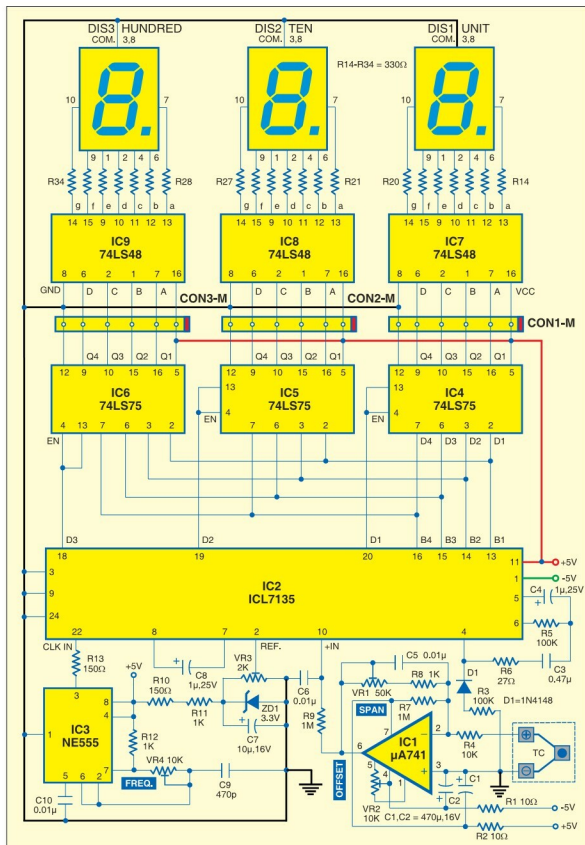


Fig. 2: Digital temperature controller using thermocouple

proximately  $42 \mu\text{V}/^\circ\text{C}$  and it is most suited for oxidising environments.

Details of thermocouple types and temperature table are included in this month's EFY-CD. See the K-type thermocouple temperature chart for

calibration of the digital temperature controller.

## Circuit description

Fig. 1 shows the block diagram of the digital temperature controller.

It comprises a signal amplifier, ADC, display unit, comparator along with relay control and power supply. A thermocouple is used as the temperature sensor. Its output is amplified by the signal amplifier and converted into a digital signal using the ADC. This digital data is shown on a 7-segment display (with the help of the display driver) and simultaneously compared with the preset temperature data. When the temperature (digital data) crosses the preset temperature data, the relay energises to control the heating element of the oven.

Fig. 2 shows the circuit of the digital temperature controller using thermocouple. At the heart of the circuit is the signal amplifier along with the ADC.

**Signal amplifier.** The signal amplifier is nothing but the thermocouple voltage amplifier. Linear operational amplifier  $\mu\text{A}741$  (IC1) is used for thermocouple voltage amplification. The input signal for the amplifier (approximately  $42 \mu\text{V}/^\circ\text{C}$ ) is generated by thermal effects of the thermocouple. The thermocouple-generated microvolt signal is amplified to millivolt (mV) with variable-gain

adjustment by feedback preset VR1 connected from inverting input pin 2 to output pin 6 to set the required amplifying factor as per the requirement. Preset VR2 is used for offset-null (zero) adjustment. Output pin 6 of  $\mu\text{A}741$